Priyanshu Dhingra

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Website Google Scholar GitHub LinkedIn

Objective

Motivated graduate student pursuing an MSc in Applied Computing (MScAC) at the University of Toronto with over two years of combined industry and research experience spanning production LLM agents for financial applications and published work in speech and computer vision. Driven by building technology from the ground up and turning ideas into products that can make a meaningful impact.

Education

University of Toronto **September 2026 – December 2027 (expected)**
Master of Science in Applied Computing (AI Concentration)
Coursework (Ongoing): Neural Networks and Deep Learning, Imitation Learning for Robotics

Rajiv Gandhi Institute of Petroleum Technology, Rae Bareli **December 2021 – May 2025**
Bachelor of Technology in Information Technology CPI: 9.36/10.00

Skills

Languages	Python, SQL, C++
ML & Modeling	PyTorch, Transformers, Scikit-Learn, Model Training, Evaluation, Inference
Agentic Systems	LLM Agents, Multi-Agent orchestration, LangGraph, LangChain, RAG, Vector stores, ChromaDB
Backend & APIs	FastAPI, Flask, REST APIs, OpenAPI, Swagger, Postman
Engineering Tools	Docker, Git, Pytest, Weights & Biases, Monitoring & Debugging, Deployment Automation

Experience

Teciem (formerly Finastra - Treasury & Capital Markets BU) **January 2025 – July 2026**
AI Software Engineer | *Project: Kondor AI Assistant*

- Developed a **dependency schema agent** for Kondor Assistant, a production **multi-agent** financial assistant, using graph-based join retrieval with **LangGraph, LangChain, and Azure OpenAI**, cutting LLM tokens per query by **30%**.
- Implemented **access controls on AI-generated SQL**, achieving **100% enforcement of user-specific data-access permissions** and preventing unauthorized access to financial data.
- Replaced LLM-generated trade and event hyperlinks with a **deterministic pipeline** using deal-ID extraction and **SQL** eligibility checks, reducing hyperlink test failures by approximately **90%**.
- Built an **LLM-based evaluation-data pipeline** using **LangChain and Azure OpenAI** to convert **2,000+ QA business cases** into structured JSON test datasets, reducing manual preparation effort by approximately **95%**.
- Developed automated evaluations across **2,000+ QA cases**, measuring **96% response accuracy**, and regression tests for prohibited database writes, out-of-scope requests, and event-link eligibility.

Carnegie Mellon University **September 2024 – January 2025**
Computer Vision Research Intern | *Project: Vision transformer for 3D Subtomogram Alignment*

- Co-authored **BOE-ViT**, a **vision transformer** framework for self-supervised **3D subtomogram alignment** published at **IEEE/CVF CVPR 2025**, contributing to implementation and experiments.
- Implemented **Fourier-domain missing-wedge masking** and investigated interpolation-based correction to address incomplete angular sampling in cryo-electron tomography.
- Achieved **77% lower rotation error** and **62% lower translation error** than Gum-Net at **SNR 0.01** as part of the BOE-ViT team, contributing to **Polyshift and MARE** implementation and experiments.
- Built **synthetic subtomogram generation** and **2D-slice visualization** pipelines supporting evaluation across **5 macromolecular structures** and **4 noise levels**, and debugged **PyTorch** training workflows.

Nanyang Technological University **March 2024 – August 2024**
Speech & Language AI Intern | *Project: Automatic Speech De-identification Systems*

- Developed an **Entity-Aware ASR** model by fine-tuning **Whisper** to jointly transcribe **speech** and identify personally identifiable information (PII), achieving a **PII detection F1 of 0.79** with synthetic-data augmentation.
- Developed an **LLM-based synthetic-data pipeline** paired with **WhisperSpeech TTS** to generate annotated speech-text pairs, expanding the text corpus from **992 to 3,592 sentences**, a **262% increase**.
- Evaluated **pipeline and end-to-end approaches** across augmentation strategies on held-out speech; the best end-to-end configuration achieved **14.8% higher PII detection F1** than the best pipeline.
- First-authored research on **speech privacy and de-identification** published at **IALP 2024, TENCON 2024, and ICAICTA 2024**, and in **IJALP 2025**.

Selected Publications

- **BOE-ViT: Boosting Orientation Estimation with Equivariance in Self-Supervised 3D Subtomogram Alignment**. *IEEE/CVF CVPR, 2025*. Co-author. (Paper)
- **Leveraging Large Language Models for Speech De-Identification**. *IJALP 2025*. First author. (Paper)
- Additional first-author papers: **IALP, TENCON, ICAICTA 2024**. **Full publication list: Google Scholar**